

Sampling-noise floor of the Jensen–Shannon distance between two empirical histograms

1 Setup

Let P be a fixed distribution on K bins with PMF $p_1, \dots, p_K$ ($\sum_k p_k = 1$). Draw two independent samples D_1, D_2 of size N each from P , and let

$$\hat{p}_k = \frac{1}{N} \sum_{x \in D_1} \mathbf{1}\{x \in \text{bin } k\}, \quad \hat{q}_k = \frac{1}{N} \sum_{x \in D_2} \mathbf{1}\{x \in \text{bin } k\}$$

be the empirical PMFs. Define the midpoint $\hat{m}_k = (\hat{p}_k + \hat{q}_k)/2$ and the Jensen–Shannon divergence (in bits)

$$\text{JS}(\hat{p}, \hat{q}) = \frac{1}{2} \text{KL}(\hat{p} \parallel \hat{m}) + \frac{1}{2} \text{KL}(\hat{q} \parallel \hat{m}),$$

and the JS *distance* $d = \sqrt{\text{JS}(\hat{p}, \hat{q})} \in [0, 1]$.

Our question: under the null hypothesis that D_1 and D_2 come from the same distribution, what is $\mathbb{E}[d]$?

2 Result

Proposition 1. *For N large and the two samples drawn iid from a common distribution with K bins,*

$$\mathbb{E}[d] \approx \sqrt{\frac{K - \frac{3}{2}}{4N \ln 2}}.$$

For $N = 1000$ and $K = 30$, this gives $\sqrt{28.5/(4000 \ln 2)} \approx 0.101$, which matches the empirical floor we observe in our experiments (values in the range 0.10–0.12 across queries).

3 Proof

Step 1: quadratic expansion of JS divergence. Write $\delta_k = \hat{p}_k - \hat{q}_k$, so that $\hat{p}_k - \hat{m}_k = \delta_k/2$ and $\hat{q}_k - \hat{m}_k = -\delta_k/2$. A second-order Taylor expansion of the KL divergence around its second argument gives

$$\text{KL}(r \parallel s) = \frac{1}{\ln 2} \sum_k \left[(r_k - s_k) + \frac{(r_k - s_k)^2}{2s_k} + O((r_k - s_k)^3) \right],$$

where the linear term drops out because $\sum_k r_k = \sum_k s_k = 1$. Hence

$$\begin{aligned} \text{KL}(\hat{p} \parallel \hat{m}) + \text{KL}(\hat{q} \parallel \hat{m}) &\approx \frac{1}{2 \ln 2} \sum_k \frac{(\delta_k/2)^2 + (\delta_k/2)^2}{\hat{m}_k} \\ &= \frac{1}{4 \ln 2} \sum_k \frac{\delta_k^2}{\hat{m}_k}, \end{aligned}$$

and therefore

$$\text{JS}(\hat{p}, \hat{q}) \approx \frac{1}{8 \ln 2} \sum_k \frac{\delta_k^2}{\hat{m}_k}. \quad (1)$$

Step 2: expectation. Under the null, $\mathbb{E}[\delta_k] = 0$ and $\hat{m}_k \rightarrow p_k$ in probability. Moreover, $\hat{p}_k$ and $\hat{q}_k$ are independent, each Binomial(N, p_k)/ N , so

$$\mathbb{E}[\delta_k^2] = \text{Var}(\hat{p}_k) + \text{Var}(\hat{q}_k) = \frac{2p_k(1-p_k)}{N}.$$

Plugging into (1),

$$\mathbb{E}[\text{JS}] \approx \frac{1}{8 \ln 2} \sum_k \frac{2p_k(1-p_k)/N}{p_k} = \frac{1}{4N \ln 2} \sum_k (1-p_k) = \frac{K-1}{4N \ln 2}. \quad (2)$$

This expectation depends only on K and N — not on the shape of P .

Step 3: from divergence to distance. The normalized statistic

$$T = \frac{N}{2} \sum_k \frac{\delta_k^2}{p_k}$$

is asymptotically χ_{K-1}^2 distributed: the differences $\delta_k/\sqrt{2p_k/N}$ are approximately standard normal and satisfy the single linear constraint $\sum_k \delta_k = 0$. Combining with (1) and $\hat{m}_k \rightarrow p_k$,

$$\text{JS} \approx \frac{T}{4N \ln 2},$$

so

$$d = \sqrt{\text{JS}} \approx \frac{\sqrt{T}}{\sqrt{4N \ln 2}} \sim \frac{\sqrt{\chi_{K-1}^2}}{\sqrt{4N \ln 2}}.$$

For moderate-large K , the chi distribution concentrates with $\mathbb{E}[\sqrt{\chi_\nu^2}] \approx \sqrt{\nu - \frac{1}{2}}$, hence

$$\mathbb{E}[d] \approx \sqrt{\frac{K - \frac{3}{2}}{4N \ln 2}}.$$

□

4 Discussion

Scaling. The floor decays as $1/\sqrt{N}$ and grows as $\sqrt{K}$. Doubling N shrinks the typical JS distance by $\sqrt{2}$; doubling the bin count inflates it by roughly $\sqrt{2}$.

Numerical values.

N	K	$\mathbb{E}[d]$
1000	20	0.083
1000	30	0.101
1000	100	0.188
10000	30	0.032

Practical implication. A JS distance near the value in Proposition 1 is consistent with the two samples coming from the same distribution; nothing can be concluded from it. A useful per-query test is the two-sample Kolmogorov–Smirnov statistic on the raw samples (no binning), or a permutation test on Wasserstein distance.